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Intrinsic differences in dispersal between populations of gastropods separated by a few metres: Evidence from reciprocal experimental transplantation

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Abstract

Differences between populations in dispersive behaviour can strongly influence population and community structure and have important implications for evolution. Differences in dispersal can be caused by intrinsic differences among populations or by different extrinsic cues. In a semi exposed shore in Co. Wicklow, Ireland, intertidal gastropods, *Littorina littorea* moved long distances in areas with low natural population density compared to areas with high natural density but were not influenced by density per se. A reciprocal experimental transplantation distinguished between intrinsic and extrinsic factors as causes of differences in dispersive behaviour. Snails from the area of high density were transplanted to the area of low density and vice versa. Comparisons were also carried out with controls (disturbed and undisturbed individuals) in each area. Dispersal of transplanted snails was compared with that of snails translocated within each area. Mean distances displaced and percentage dispersal were monitored after two days period. To test temporal generality the experiment was done twice. The two experiments yielded different outcomes. The majority of evidence supported the intrinsic model: transplanted snails dispersed differently from controls in their new area and similarly to controls in their area of origin. However, in one of the experiments there was some evidence suggesting influence of extrinsic factors or an interaction between extrinsic and intrinsic factors. Further research is required to identify whether the observed intrinsic differences are genetic or due to internal conditions that have been modified by the animal's present and/or past environment. This study reinforces the value of repeated experimental transplantation to characterize factors causing differences in behaviour.
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1. Introduction

Dispersal is one class of behavioural characteristics, which can vary between local populations of a species (Crowe, 1996a; Crowe and Underwood, 1998; Coltman, 2005). It can strongly influence population and community structure (Davidson et al., 2005; Finn and Poff, 2005) and has important implications for evolution. The study of evolution has focused primarily on the two fundamental determinants of fitness: reproductive rates and survival probabilities. Moving from one location, or staying behind, can strongly affect both of these parameters (Postma and van Noordwijk, 2005). Ecologists have often assumed that spatial variation in

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density reflect differences in various aspects of habitat quality (e.g. amount of food, number of predators, presence of mates) that influence an organism's probability of moving from one location, or staying behind (Parker and Stuart, 1976). Considerable research has gone into identifying causes and consequences of dispersal in different species and the relative importance of different factors affecting dispersal (e.g. Crowe, 1996a; Stamps et al., 2005; Chapple and Keogh, 2005; Forward et al., 2005).

Generally mobile species respond to high competition by emerging from areas of high density to areas of lower density where there is less pressure on resources. The effects of density and/or food limitation are well known for a number of insects, especially sap-feeding insects as aphids and plant-hoppers (Dixon, 1985; Denno et al., 1994; Denno and Peterson, 1995) and also for a range of other invertebrates such as collembolans, grasshoppers, true bugs, moths, beetles and flies (e.g. Johnson, 1969; Bengtsson et al., 1994; Denno and Peterson, 1995; Fonseca and Hart, 1996; Applebaum and Heifetz, 1999). Decreases in micro-algal food have been shown to increase dispersal in a number of intertidal gastropods as well (Branch and Branch, 1981; Mackay and Underwood, 1977).

Littorina littorea is an intertidal marine gastropod, which is principally a grazer of epilithic microalgae (Hawkins and Hartnoll, 1983; Konan et al., 1992). It is harvested in many parts of its range (e.g. Cummins et al., 2000) and can exert significant control over algal assemblages (e.g. Bertness, 1984; Lubchenco, 1983). In its benthic phase, it exhibits considerable variation in density over small spatial scales and disperses actively through a range of microhabitats on rocky shores. It is therefore a tractable model organism for investigating processes affecting dispersal. In a previous study (Rajasekharan et al., communicated), *L. littorea* was shown to move long distances in an area with low natural population density compared to an area with high natural density. Dispersal from plots of a range of experimental densities in the two areas were also found to be similar but dispersal differed between the two areas, indicating that dispersal was influenced by factors other than population density.

Three models could explain the observed differences in the behaviour of *L. littorea* (Table 1). The first is that there are intrinsic differences in the *L. littorea* that occupy the two areas, causing them to behave differentially. The second is that there are no intrinsic differences in *L. littorea* occupying the two areas but that they respond to a different set of extrinsic environmental cues in each area and therefore behave differentially. The final model is that some combination of intrinsic and extrinsic factors acts to produce the observed pattern of behaviour.

We hypothesized that, if there are intrinsic differences in snails between the two areas of the shore; snails transplanted from the area with high natural density to the area with low natural density should retain the behavioural characteristics of the area of high natural

Table 1
Summary of models, hypotheses and tests to investigate differential dispersal of *L. littorea* from an area of high density when compared to dispersal from an area of low density

Models	Hypothesis	Test
(1) Intrinsic factors: intrinsic differences between populations of *L. littorea* in two parts of the shore cause them to behave differently	(1) Transplanted snails will retain behaviour characteristic of their area of origin and will disperse (i) differently from controls in their new area and (ii) similarly to controls in their area of origin	To distinguish among hypotheses 1 to 3, transplant snails from the area of high density to the area of low density (and vice versa) and compare dispersal of transplanted snails to that of control individuals manipulated within their respective areas of origin. In addition, compare dispersal of transplanted snails originating from an area with that of controls manipulated with in it
(2) Extrinsic factors: different cues in each area cause them to behave differentially	(2) Transplanted snails will alter their behaviour in response to cues in the new area and will disperse (i) similarly to controls in their new area and (ii) differently from controls in their area of origin	
(3) Interaction of intrinsic and extrinsic factors: differences in behaviour in two areas are caused by an interaction or combination of intrinsic differences in *L. littorea* and differences in environmental cues	(3) Either, snails transplanted from one of the areas will behave as predicted in hypothesis 1 and snails transplanted from other areas will behave as predicted in hypothesis 2 or behaviour of transplanted snails become intermediate between the behaviours in the area of origin and that to which they were taken	

density (Hypothesis 1; Table 1). If there are no intrinsic differences between populations, transplanted snails should alter their behaviour in response to extrinsic cues in the new area (Hypothesis 2; Table 1). If an interaction of these two factors is responsible for the observed differences, the outcome would either vary from area to area or the behaviour of snails would become intermediate between that typical of snails in their new area and that of snails in their area of origin (Hypothesis 3; Table 1). Crowe and Underwood (1999) demonstrated the need for repeated experimental transplantations to understand differences in dispersal in an intertidal gastropod, *Bembicium auratum*. Outcomes varied considerably in space and time in mangrove forests and rocky shores. Therefore in the current study, the experiments were done twice.

To distinguish between the effects of the new environment on the behaviour of snails and the effect of simply being moved to a new area, it is necessary to translocate snails to another area within the habitat of origin ('translocation control' — Chapman, 1986; Crowe and Underwood, 1999). A comparison of these two treatments will provide a direct test of the hypothesis. To be able to interpret the results in terms of the behaviour of unmanipulated snails, it is also necessary to incorporate controls to identify any artifacts of these manipulations. To do this, it is necessary to compare behaviour of manipulated snails with that of disturbed animals (disturbance control) and unmanipulated animals (undisturbed control). Any differences between experimental and disturbed/undisturbed snails would be attributable to aspects of the experimental manipulation (Chapman, 1986, 1999; Crowe, 1996b; Crowe and Underwood, 1999).

2. Methods

The reciprocal experimental transplantations were carried out at Bray Head, Co. Wicklow, a moderately exposed rocky shore on the east coast of Ireland. Topography of the two areas of Bray Head consisted of small rough boulders, and rocks and smooth stones. The experiment was run in two 15 × 15 m areas at mid-shore level separated by 20 m. Density of *L. littorea* varied from 43 to 827 per 0.25 m^2 in the area with high natural density. In the area with low natural density, *L. littorea* density ranged from 12 to 123 per 0.25 m^2. The experiment was done twice: Experiment 1 in June 2003 and Experiment 2 in August 2003.

At each area, 16 plots of size 0.5 × 0.5 m were selected and marked. Plots at each area were randomly assigned to each of the following treatments: (1) transplantation, (2) translocation control, (3) disturbance control and (4) undisturbed control. Twenty snails were collected from each plot. It has been reported previously that age is an important criterion in predicting rates of movement of gastropods (Mackay and Underwood, 1977; Shepherd, 1986; Crowe, 1996a). Therefore, individuals with shell height <15 mm were excluded from consideration. In Experiment 1, in three plots in the area of low density (2 undisturbed control, 1 disturbed control) only between 13 and 15 *L. littorea* of appropriate size were found. In experiment 2, in five plots (3 undisturbed control, 2 disturbed control — all in the area of low natural density) only between 11 and 17 *L. littorea* of appropriate size were found. For these plots, no additional *L. littorea* were collected and only the number present was used. This prevents the problems associated with altering natural densities and introducing strange animals (Underwood, 1986). The shells of all the selected snails were painted with a colour specific to that plot and easily distinguished from colours of nearby plots. The snails were then placed haphazardly, but uniformly in the center of assigned plots. (Hereafter, subscript HD and LD indicate the areas to which snails were moved or placed; HD indicating the area of high density and LD indicating the area of low density). Snails to be transplanted were moved to their assigned destination plots [those in high natural density area moved to plots in low natural density area (TP_{LD}) and those in low natural density area moved to plots in high natural density area (TP_{HD})]. Snails allocated to translocation were moved to plots in the same area of the shore from which they had been collected (Translocation in the area of high natural density — TL_{HD}; Translocation in the area of low natural density — TL_{LD}). Snails assigned to disturbance treatment were replaced on the plot from which they had been taken (Disturbance control in the area of high natural density — D_{HD}; Disturbance control in the area of low natural density — D_{LD}). For the undisturbed control, natural behaviour of snails was monitored (Undisturbed control in the area of high natural density — U_{HD} and Undisturbed control in the area of low natural density — U_{LD}). As it was not possible to keep track of unmarked snails, undisturbed control was done by painting snails *in situ* without picking them up from the plots.

Four replicates were run for each treatment. One replicate from each treatment was set up per day and the data were collected after 2 days. The number of snails still in the plot was recorded and a search was made for individuals that had dispersed. For those that were found, the distance they were from the center of the plot was recorded. Clearly the measurement was not the

distance the snails had actually moved, as it took no account of the path they took. It was therefore referred as the 'displaced distance' of the snails (Underwood and Chapman, 1992; Crowe, 1996a; Crowe and Underwood, 1999). Percentage dispersal was also calculated by dividing the number of *L. littorea* dispersed from the plot by the initial number in the plot and multiplying by 100.

Two types of analyses were carried out to distinguish between the alternative hypotheses: analysis by 'destination' and analysis by 'origin'. First, the behaviour of snails transplanted to an area was compared with the behaviour of those manipulated within it (analysis by destination). In addition, the behaviour of transplanted snails originating in an area was compared with that of those manipulated with in it (analysis by origin). Displaced distances and percentage dispersal were analysed in each of these ways using analysis of variance (ANOVA). Treatments were analysed as a fixed factor and plots as a random factor. For those terms that were found to be significant, means were compared using Student–Newman–Keuls (SNK) test (Underwood, 1997). Analyses of displaced distances were necessarily a subset of all the available data (data were randomly discarded so that the design would be balanced). A regression was done of the mean values from the randomly chosen subset and the mean values from the full data sets to estimate if some of the means estimated by the subsets of data were very different from the means estimated by all the data (Crowe and Underwood, 1999). Ninety-eight percent of the variation in the estimates using subsets of data was accounted for by estimates using all available data and there were no strong outliers.

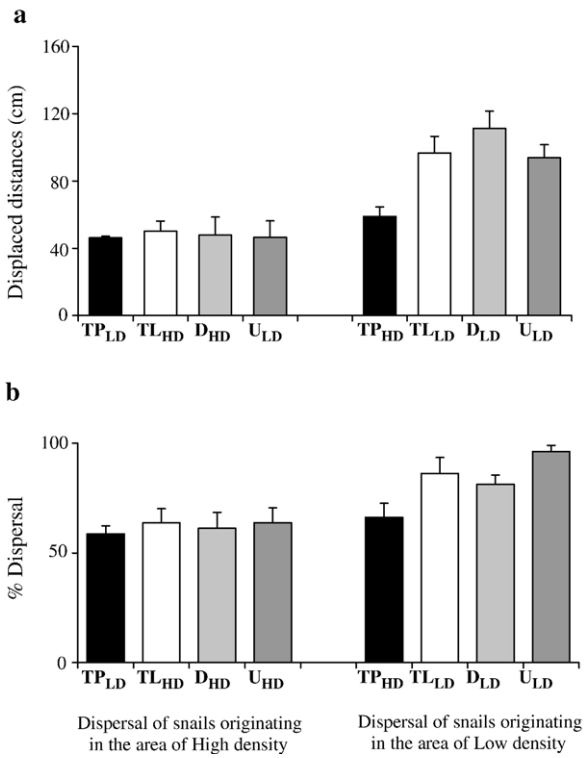

Fig. 1. Experiment 1, June 2003. (a) Displaced distances, and (b) percentage dispersal of *L. littorea* from plots after two days. Bars grouped by origin: TP_{LD} — transplanted to the area of low density from the area of high density; TL_{HD} — translocated within the area of high density; D_{HD} — disturbance control in the area of high density; U_{HD} — undisturbed control in the area of high density; TP_{HD} — transplanted to the area of high density from the area of low density; TL_{LD} — translocated within the area of low density; D_{LD} — disturbance control in the area of low density; U_{LD} — undisturbed control in the area of low density; $n=4$ plots. Shown are mean + S.E.

3. Results

3.1. Experiment 1: June 2003

Overall, percentage recapture of snails in all the treatments in the area of high density varied from 55 to 80%. In the area of low-density, recapture varied from 50 to 75%.

In the analysis of displaced distances 'by origin', snails transplanted to the area of low density from the area of high density dispersed similar distance to those translocated within the area of high density (Support for Model 1; Fig. 1a; Table 2a(i); $TP_{LD}=TL_{HD}=D_{HD}=U_{HD}$; SNK test, $P>0.05$). i.e. transplanted snails retained behaviour typical of snails in their area of origin and unlike that of snails to which they had been taken. Snails transplanted from the area of low density to the area of high density, on the other hand, dispersed significantly shorter distances compared to the snails translocated within the area of low density (Support for Model 2; Fig. 1a; Table 2a(ii); $TP_{HD}<TL_{LD}=D_{LD}=U_{LD}$, SNK test, $P<0.01$).

When displaced distances were analysed 'by destination', snails transplanted to the area of high density from the area of low density dispersed significantly longer distances compared to the control snails in the area of high density (Fig. 1a; Table 3a(i); $TP_{HD}>D_{HD}=U_{HD}=TL_{HD}$, SNK test, $P<0.001$). Those transplanted to the area of low density, dispersed significantly shorter distances compared to the controls in the area of high density (Support for Model 1; Fig. 1a; Table 3a(ii); $TP_{LD}<TL_{LD}=D_{LD}=U_{LD}$, SNK test, $P<0.05$).

There was a smaller percentage dispersal from plots by snails, which had been transplanted to the area of high density compared to snails translocated within the

Table 2
Analyses by origin: June 2003 — (a) analysis of distances displaced by *L. littorea* originating from plots in (i) the area of high-density, (ii) area of low-density; (b) analysis of % dispersal of *L. littorea* originating from plots in (i) the area of high-density, (ii) area of low density

	(i)			(ii)		
Source of variation	*df*	MS	*F*	*df*	MS	*F*
(a)						
Treatment=*T*	3	0.20	0.03	3	5.81	12.69**
Plot=*P*	3	2.11	1.21	3	0.88	1.79
T×*P*	3	6.06	3.48**	3	0.45	0.93
Residual	176	1.74		176	0.49	
(b)						
Treatment=*T*	3	22.91	0.20	3	625.00	6.98**
Residual	12	114.58		12	89.58	

(a) Data are transformed, $\chi' = \mathrm{Log}_e(\chi+1)$; Cochran's test: ns; $n=12$ snails.
(b) Data not transformed, Cochran's test: ns; $n=4$ plots.
**Denotes significance at $P<0.01$.

area of low density (Support for Model 1; Fig. 1b; Table 2b(ii); $TP_{HD}<TL_{HD}=D_{HD}=U_{HD}$, SNK test, $P<0.01$). On the other hand, percentage dispersal of snails transplanted to the area of low density from the area of high density was equivalent to that of control snails within the area of high density (Support for Model 3; Fig. 1b; Table 2b(i); $TP_{LD}=TL_{HD}=D_{HD}=U_{HD}$; SNK test, $P>0.05$).

When the percentage dispersal of the snails was analysed 'by destination' snails transplanted to the area of high density from the area of low density were shown to behave like those manipulated within the area of high density (Support for Model 2; Fig. 1b; Table 3b(i); $TP_{HD}=TL_{HD}=D_{HD}=U_{HD}$, SNK test, $P>0.05$). In the area of low density, on the other hand, snails transplanted to the area of low density from the area of high density dispersed in smaller numbers compared to the snails that were translocated within the area of high density (Support for Model 1; Fig. 1b; Table 3b(ii); $TP_{LD}<TL_{LD}=D_{LD}=U_{LD}$, SNK test, $P<0.01$).

Table 3
Analyses by destination: June 2003 — (a) analysis of distances displaced by *L. littorea* originating from plots in (i) the area of high-density, (ii) the area of low-density; (b) analysis of % dispersal of *L. littorea* originating from plots at (i) area of high-density, (ii) area of low density

	(i)			(ii)		
Source of variation	*df*	MS	*F*	*df*	MS	*F*
(a)						
Treatment=*T*	3	11.67***	50.26	3	15.64	5.94*
Plot=*P*	3	0.41	0.47	3	3.27	4.22**
T×*P*	3	0.23	0.26	3	2.63	3.39***
Residual	176	0.87		176	0.77	
(b)						
Treatment=*T*	3	16.66	0.12	3	1006.25	14.64**
Residual	12	135.41		12	68.75	

(a) Data are transformed, $\chi' = \mathrm{Log}_e(\chi+1)$; Cochran's test: ns; $n=12$ snails.
(b) Data not transformed, Cochran's test: ns; $n=4$ plots.
*Denotes significance at $P<0.05$; **denotes significance at $P<0.01$; ***denotes significance at $P<0.001$.

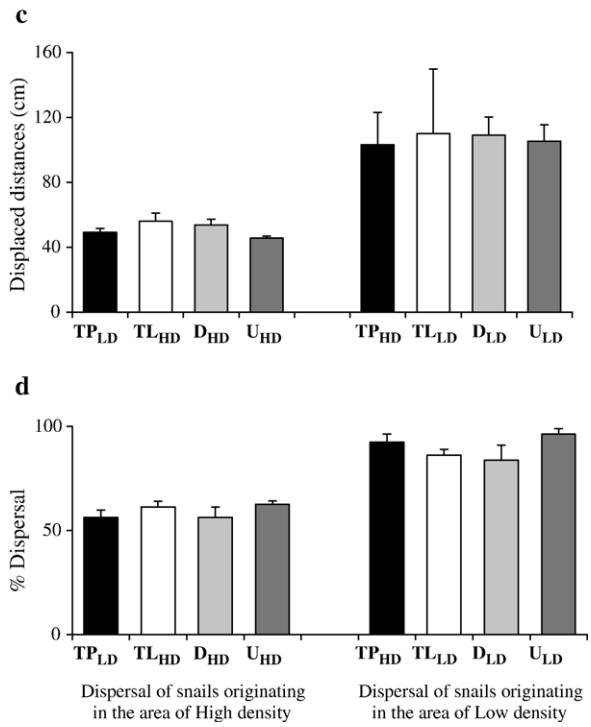

Fig. 2. Experiment 2, August 2003. (c) Displaced distances, and (d) percentage dispersal of *L. littorea* from plots after two days. All labels as in Fig. 1.

Table 4
Analysis by origin: August 2003 — (a) analysis of distances displaced by *L. littorea* originating from plots in (i) the area of high-density, (ii) area of low-density; (b) analysis of % dispersal of *L. littorea* originating from plots in (i) the area of high-density, (ii) area of low density

	(i)			(ii)		
Source of variation	df	MS	*F*	df	MS	*F*
(a)						
Treatment=*T*	3	0.580	2.72	3	0.225	1.11
Plot=*P*	3	0.597	0.56	3	0.170	0.40
T×*P*	9	0.213	0.20	9	0.203	0.48
Residual	176	1.073		176	0.423	
(b)						
Treatment=*T*	3	43.229	1.20	3	130.729	2.15
Residual	12	35.937		12	60.937	

(a) Data are transformed, $\chi' = \mathrm{Log}_e(\chi+1)$; Cochran's test: ns; $n=12$ snails.
(b) Data not transformed, Cochran's test: ns; $n=4$ plots.

3.2. Experiment 2: August 2003

Percentage recapture of snails in all the treatments in the area of high density varied from 60 to 80%. In the area of low-density, recapture varied from 65 to 75%, except in undisturbed controls where average recapture was only 50%. Findings provided clear support for Model 1.

Snails transplanted to the area of low density (TP_{LD}) displaced an average distance of 49 cm compared to translocated snails within the area of low-density (TL_{LD}), where the average displacement was 109 cm (Fig. 2c). Average displaced distances from other treatments were as follows, TP_{HD}=103 cm, TL_{HD}=55 cm, D_{HD}=53 cm, D_{LD}=108 cm, U_{HD}=45 cm, U_{LD}=105 cm (Fig. 2c). When analyses of displaced distances were carried out by origin, no significant differences were observed between treatments in the area of high density (Support for Model 1; $TP_{LD}=TL_{HD}=D_{HD}=U_{HD}$; $P>0.05$; Table 4a (i)) or in the area of low density (Support for Model 1; $TP_{HD}=TL_{LD}=D_{LD}=U_{LD}$; $P>0.05$; Table 4a(ii)).

After two days, percentage dispersal was 50–55% from TP_{LD} compared to 85–100% from TP_{HD} (Fig. 2d). Analyses of percentage dispersal by origin showed similar dispersal from all treatments in the area of high natural density (Support for Model 1; $TP_{LD}=TL_{HD}=D_{HD}=U_{HD}\approx 56\%$; $P>0.05$; Table 4b (i)) and in the area of low natural density (Support for Model 1; $TP_{HD}=TL_{LD}=D_{LD}=U_{LD}\approx 89\%$ $P>0.05$; Table 4b(ii)).

When analyses of displaced distances was carried out by destination, significant differences were found in displaced distances between snails transplanted to area of high density (TP_{HD}) and snails translocated within the area of high density (TL_{HD}) (Support for Model 1; Table 5a(i); $TP_{HD}>TL_{HD}=D_{HD}=U_{HD}$, $P<0.01$). Similar results were obtained when snails transplanted to area of low density (TP_{LD}) were compared with those translocated and within area of low density (TL_{LD}) (Table 5a(ii); $TP_{LD}>TL_{LD}=D_{LD}=U_{LD}$; $P<0.01$).

Analyses of percentage dispersal by destination revealed significant differences between treatments in the area of high density (Support for Model 1; Table 5b (i), $TP_{HD}>TL_{HD}=D_{HD}=U_{HD}$, $P<0.01$) and in the area of low density (Support for Model 1; Table 5b(ii); $TP_{LD}>TL_{LD}=D_{LD}=U_{LD}$; $P<0.01$).

Table 5
Analysis by destination: August 2003 — (a) analysis of distances displaced by *L. littorea* from plots in (i) the area of high-density, (ii) area of low-density; (b) analysis of % dispersal of *L. littorea* from plots in (i) the area of high-density, (ii) area of low density

	(i)			(ii)		
Source of variation	*df*	MS	*F*	*df*	MS	*F*
(a)						
Treatment=*T*	3	11.673	50.26***	3	14.275	50.13***
Plot=*P*	3	0.412	0.47	3	0.051	0.08
T×P	9	0.232	0.26	9	0.284	0.46
Residual	176	0.877		176	0.619	
(b)						
Treatment=*T*	3	3512.5	41.63***	3	3518.75	19.41***
Residual	12	337.5		12	725.00	

(a) Data are transformed, $\chi'=\mathrm{Log}_e(\chi+1)$; Cochran's test: ns; $n=12$ snails.
(b) Data not transformed, Cochran's test: ns; $n=4$ plots.
***Denotes significance at $P<0.001$.

4. Discussion

The majority of evidence supported the intrinsic model. However, in one of the experiments (experiment 1, June 2003) there was evidence suggesting some influence of extrinsic factors and a combined role of extrinsic and intrinsic factors. In that experiment, snails transplanted to the area of high density exhibited increased displaced distances than controls that remained in the area of high density ($TP_{HD}>D_{HD}=U_{HD}=TL_{HD}$), but the reverse was not true. Snails transplanted to the area of low density dispersed at rates similar to those of snails translocated within the area of high density and less than those of snails translocated or disturbed or left undisturbed within the area of low density ($TP_{LD}=TL_{HD}<D_{LD}$). This suggests that something intrinsic to snails inhabiting the area of high density caused them to disperse at lesser rates regardless of the area they were in. In contrast, snails from the area of low density responded to cues that prompted them to disperse greatly in the area of high density. Results from the analyses of percentage dispersal from the experimental plots revealed a different story. Percentage dispersal of the snails transplanted to the area of high density from the area of low density was similar to that of snails translocated within the area of high density ($TP_{HD}=TL_{HD}$). Thus, intrinsic factors are definitely shown to play a role in the behaviour of *L. littorea*. However, extrinsic factors may play an additional role at times.

The outcome of experiment 2 (August 2003) pointed clearly to intrinsic differences between the snails in the two area of the shore. *Littorina littorea* in the area with high density dispersed in smaller proportions and over shorter distances compared to those in the area with low natural density. *L. littorea* transplanted from the area of low density to the area of high density displaced long distances similar to snails translocated within the area of low density ($TP_{HD}=TL_{LD}$). Similarly, snails transplanted

from the area of high density to the area of low density, dispersed in smaller numbers and shorter distances like snails translocated within the area of high density ($TP_{LD}=TL_{HD}$). Displaced distances and percentage dispersal of transplanted snails were similar those of controls in their area of origin and different from controls in their new area. i.e., transplanted snails retained behavioural characteristics of their area of origin. Hence, the results supported Model 1, that there are intrinsic differences between *L. littorea* occurring in the area with high density and those in the area with low density.

Placing animals in a new environment causes the possibility of physical, chemical, biological or even social shock and adequate acclimation period is necessary to allow animals to stabilize to their new environment (Sanina and Kostetsky, 2002; Gherardi and Atema, 2005; Elekonich and Roberts, 2005). For example, at first Hobday (1995) observed no changes in the behaviour of limpets when they were transplanted within and between areas of differing wave exposure. After an acclimation period of five days, transplanted animals behaved similarly to controls moved within the area to which the transplanted animals had been moved. But no controls to assess the effect of the experimental disturbance on the behaviour of the limpets were carried out, making it difficult to draw unambiguous conclusions (Underwood, 1986, 1988). Similarly, Truscott et al. (1995) observed an immediate change in the behaviour of pond snails when exposed to aluminum and lead. As the snails became acclimated, their behaviour became similar to that of controls that had never been exposed to the metals. In our study, dispersal of transplanted *L. littorea* was monitored after only two days in the two areas. There is a possibility that the acclimation period of *L. littorina* is more than two days and the behaviour of snails would be different if dispersal was monitored for a longer period of time. To distinguish between these alternatives, dispersal experiments could be carried out for a longer period to compare the behaviour of snails that had been freshly transplanted and those that are left to acclimate in the new area.

Using reciprocal experimental transplantation, Crowe and Underwood (1999) tested the behaviour of intertidal gastropod *Bembicium auratum* in rocky shores and in mangrove forests. Outcomes of the experiments varied considerably in space and time, but there was no evidence of intrinsic differences between populations of snails in the two habitats. Some results suggested that a combination of intrinsic and extrinsic factors caused the observed differences in behaviour. Most results were consistent with the model that differences in behaviour of snails on rocky shores and from that shown in mangrove forests are caused by differences between the two habitats and not the snails. This is in contrast to findings of our study, where the majority of evidence supported intrinsic differences between populations as the cause of differential dispersal.

Broadly, there can be two possible explanations for the observed intrinsic differences in the behaviour of snails in our study; differences in the genetic make up of snails inhabiting the two areas or differences in the internal conditions of snails inhabiting the two areas. Here, by internal conditions we refer to temporary or persistent measurable behavioural, morphological and physiological traits (i.e., the phenotype) that have been modified by the organism's present and past environment, rather than differences produced by genotype. Or else a combination of genetic and phenotypic processes can cause individual differences in the tendency to disperse.

Whether or not two separate populations of a species become genetically different depends largely on gene flow (Bohonak, 1999). Classical population genetics theory predicts that, populations that frequently exchange individuals through dispersal will remain genetically similar (Palumbi, 1995). Disconnected populations, by contrast, have a greater capacity to become distinct through forces such as genetic drift and adaptation to local conditions. Mollusc species with extended larval planktonic phases are generally thought to disperse further and have more gene flow, larger geographic ranges and lower levels of genetic differentiation within populations (Waples, 1987; Williams and Benzie, 1993; Palumbi, 1995; Hohenlohe, 2004). Whereas those with direct development show high levels of genetic variability even at small spatial scales (e.g. Wilson and Gosling, 1998). *L. littorea* has a long-lived planktonic larval stage (4–7 weeks), which travel large distances (Johannesson, 1992; Reid, 1996) and are therefore expected to be fairly genetically alike at small spatial scales. Allozyme studies generally confirm this assumption in *L. littorea* (Wilkins and O'Regan, 1980; Johannesson, 1992).

Two new independent studies of wild great tits, *Parus major*, challenge the assumption that populations that frequently exchange individuals through dispersal will remain genetically similar (Coltman, 2005; Postma and van Noordwijk, 2005). Differing levels of gene flow have maintained large genetic differences at a very fine spatial scale, relative to the bird's dispersal capability. Similarly, using quantitative genetic analysis Garant et al. (2005) revealed genetic variation between populations that are separated only by 2 km. They suggest that temporal and spatial variation in bird density provide this mechanism. On average, local density of great

tits is twice as high in the east than north, and hence produced greater pressures on habitat in the east. This directed individuals that are genetically predisposed to be larger and heavier to preferentially settle in lower-density habitat in the north. The findings shed new light on the mechanisms of microevolution, because there is no reason to suspect that these phenomena are unique to great tits. The ability to detect such phenomena depends on the availability of long-term data from continuously monitored populations of marked individuals, and on the existence of biotic or abiotic gradients that may drive the underlying micro evolutionary processes. It is possible that in the current study, a similar mechanism is operating. There may be an influx of genes to the area of high-density relative to the area of low-density, thus creating genetic structuring at such a small spatial scale. Genetic characterization of these populations using DNA markers could shed light on this (Wilding et al., 2000). Previous studies based on allozyme genetic markers in *L. littorea* report genetic similarity, even between populations that are widely separated (Johannesson, 1992). Allozyme studies can be influenced by selection of the loci involved, and may require extensive time to reach a point at which separate gene pools are detectable, leading to low resolution. In contrast, more recent genetic methods based on highly polymorphic markers (e.g. microsatellite markers) and the assignment of individuals to particular populations depending on the combination of alleles that they carry have more promise, as they allow direct discrimination between immigrants and residents (Balloux and Lugon-Moulin, 2002).

Another possible basis for intrinsic differences over small distances is physiological plasticity. A few studies on marine gastropod species with contrasting abilities for larval dispersal have shown that planktonic developers may not only be more genetically homogeneous than direct developers but may also be physiologically homogeneous (Levins, 1968; Gooch, 1975; Gallardo and Perron, 1982; Janson, 1987). Studies by Parsons (1997, 1998), on the other hand, pointed to high levels of phenotypic variation through physiological plasticity in species with wide dispersal. She studied the role of dispersal ability in determining phenotypic differentiation and plasticity of growth in two marine gastropods; *Austrocochlea constricata*, a planktonic developer and *Bembicium vittatum*, a direct developer. Despite wider dispersal, high levels of phenotypic plasticity were observed among populations of *A. constricata* while genetic influences on growth were greatest in *B. vittatum.* The results thus suggested that wide dispersal enhances physiological plasticity. In our study intrinsic differences in *L. littorea* between populations in the two areas can be potentially ascribed to phenotypic differentiation due to physiological plasticity.

Internal conditions of animal may also explain the observed intrinsic differences in behaviour by snails in the two areas. For example, in several species, parasitism is shown to alter the fitness of host and dispersal pattern of the host (Boulinier et al., 2001). *L. littorea* is an intermediate host of a number of diagenetic parasites (Hughes and Answer, 1982; Lambert and Farley, 1968; Huxham et al., 1993). It has been reported that level of parasitic infection can change behaviour of this species (Williams and Ellis, 1975). Similarly, trematode infection is shown to limit the range of movement of parasitized trochid snail *Diloma subrostrata*, contributing to spatial structuring (Miller and Poulin, 2001). It may be that parasites are more prevalent in the area of high natural density, than in the area of low density leading to higher dispersal in the low-density area. Surveys of parasite abundance in the two areas could give an estimate of the mean prevalence (i.e. percentage of infected individuals; Bush et al., 1997) as an initial step in examining this model.

A study on *Littorina saxatilis* (Olivi) reported sex and reproductive condition as factors causing significant differences in magnitude of movements within a population (Pardo and Johnson, 2004). Gravid females moved on average, shorter distances during foraging excursions than the other groups (non-gravid females and males). Reduction of movement in brooding individuals is also well known in some terrestrial species (Clutton-Brock, 1991; Trumbo, 1996) and in a number of aquatic vertebrates, including amphibians and fish (Smith and Wooton, 1995; Crump, 1996), and freshwater invertebrates (Smith, 1976). *L. littorea* has been reported to spawn at various times on coasts of British Isles from seasonal to year round (Smith and Newell, 1955). The observed temporal differences in the outcome of experiments could be potentially linked to the reproductive stage of *L. littorea* at different times.

The results of our study suggest that changes of behaviour can occur between populations even if they are separated only by a few metres. Hence, models based on detailed knowledge of the behaviour of *L. littorea* in one area do not necessarily reflect behaviour in another area. However, our study was not designed to characterise differences between populations in the two areas of the shore in terms of sex, reproductive stage and genetic or phenotypic variation. Future studies should consider the effect of these potential factors affecting dispersal. Variation in the outcomes of the experiment reinforces the need for repeated experimental transplantations (see Crowe and Underwood, 1999).

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